Cell Division: Meiosis

What is the definition of meiosis?

Meiosis is

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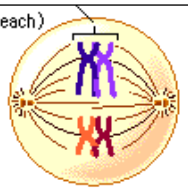
What is the purpose of meiosis?

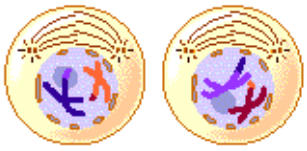
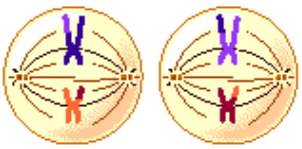
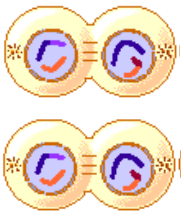
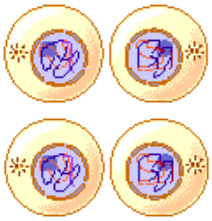
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Where does meiosis occur?

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What are the stages involved in meiosis?

Phase of Meiosis	Description of Phase	Images of Phase
Interphase (before meiosis)	The cell functions as normal The DNA is in the form of threadlike chromatin The DNA duplicates itself	
Prophase I	DNA condenses into visible chromosomes Nuclear membrane breaks down and spindle forms **Homologous chromosomes pair up and crossing over occurs**	
Metaphase I	**Paired chromosomes line up in the centre of the cell** Spindle fibre attaches to the centromeres	
Anaphase I	The spindle fibres contract, pulling one of each pair of chromosome to opposite ends of the cell	

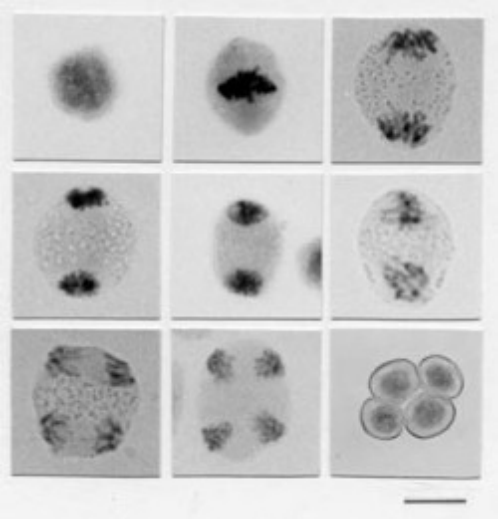
Telophase I	New nuclear membranes start to form Cell pinches in the middle and splits into two	
Prophase II	Nuclear membrane breaks down Spindle forms around the chromosomes	
Metaphase II	Chromosomes line up in the centre of the cells Spindle fibres attach to centromeres	
Anaphase II	Chromatids separate and move to opposite poles of the cell	
Telophase II	New nuclear membranes form Cells divide into two **Four daughter cells are produced with half the number of chromosomes**	
Interphase (after meiosis)	DNA unwinds into threadlike chromatin	

Crossing Over

This is a process that is unique to meiosis

Parts of adjacent chromosomes are able to cross over and exchange parts of their DNA

This creates four distinct chromatids





Interphase



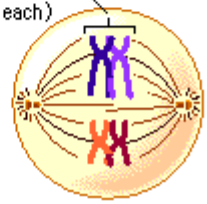
MEIOSIS I

Prophase I
Synapsis and crossing over occur.



Tetrad (paired homologous chromosomes with two chromatids each)

Metaphase I
Tetrads line up on the metaphase plate.



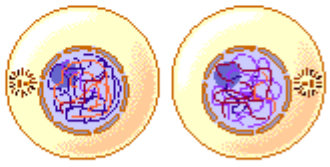
Anaphase I
Homologous pairs separate.



Telophase I



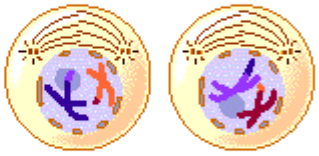
Cytokinesis I



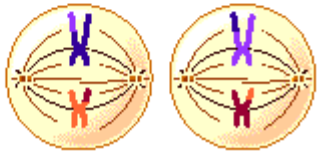
To Prophase II

MEIOSIS II

Prophase II



Metaphase II
Chromosomes line up on the metaphase plate.



Anaphase II
Sister chromatids separate.



Telophase II



Cytokinesis II



4 haploid daughter cells are formed, each having only one chromosome of each homologous pair.

